

Stimulus PCs

About



Stimulus PCs (STIM-PC) are available for use within the Imaging Suites and Testing Rooms of the FIL.

These systems provide Windows environments from which experiments may be run, access is permitted with FIL IT credentials only.

Please familiarise yourself with their [Specifications](#) and [Imaging Suite configurations](#).

Imaging Suite User Guide

Use the following guides to configure FIL STIM-PCs and see the [Personal STIM-PC](#) section if using a personal device.

3T MRI

Within the 3T MRI suites a WIN11 STIM-PC is available.
The basic configuration of the system is shown below:

[3T STIM-PC User Guide](#)

Standard Configuration of 3T Stimulus PC	
1) Choose the Stimulus PC Required (<i>do not switch ON</i>)	
- WIN11 STIM-PC: Windows 11, Psychtoolbox 3.0.16, Psychopy, Internet - Personal PC: Laptop etc	
See the Stimulus PC specification table for full details	
2) Connect the 'Master USB' to the PC	
- By default, this will be connected to the WIN11 PC - Connect 'Master USB' to any USB port of the desired PC - This will connect: Mouse, Keyboard, Audio and Keypads	

Standard Configuration of 3T Stimulus PC	
3) Connect Personal PC to Projector <i>(if required)</i> - On the HDBaseT, locate the HDMI port labelled 'Connect Laptop HDMI Here' - Connect the 'Laptop HDMI Cable' to this port, connected the other end to the personal PC See the Projector page for full specifications	
4) Turn ON Projector HDBaseT Unit - Press: ON - LED will turn Green On Remote - Press: ON - Wait for projector to turn on - Epson Logo appears on small Siemens display	
5) Select Projector Input On Remote - Press: HDMI or HDBaseT WIN11 STIM-PC = HDMI Personal PC = HDBaseT	
6) Configure Scan Triggers Key Press Triggers - Trigger a script from a keypress of '5' - Plug BNC 'Keypress 5' into the 'Keypress Trigger Unit' Serial Triggers & No Triggers - Trigger a script from serial output - Unplug BNC 'Keypress 5' from 'Keypress Trigger Unit'. - <i>Personal Laptops will require a serial connection, speak to cognitive interface</i> See the MRI Scan Synchronisation page for full detail.	
7) Turn ON Stimulus PC Computer issues are reduced by connecting all devices and configuring the projector prior to boot	

Standard Configuration of 3T Stimulus PC

8) Check Display Options

Display options can be configured but must be returned to default:

- Right Click Desktop
- Click: Screen Resolution/Display Settings

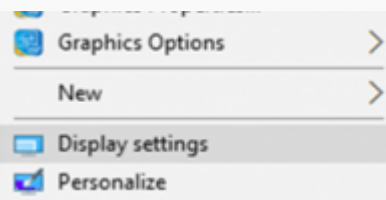
Default

Multiple displays: Extended these displays

Dell P2419H monitor : 1920×1080

(Recommended)

Epson PJ projector : 1920×1200 (Recommended)



When Finished

9) Turn OFF Projector

On Remote:

- Press: **Standby**, two times
- Wait for projector to turn OFF



10) Return To Default

- Revert any display settings back to default
- Reconnect any cables back to the WIN11 STIM-PC

11) Shutdown PC

7T MRI

Within the 7T MRI suite a WIN11 STIM-PC is available for use.

Basic configuration of this system can be seen below:

7T STIM-PC User Guide

Standard Configuration of 7T Stimulus PC

1) Choose the Stimulus PC Required (*do not switch ON*)

- **WIN11 STIM-PC:** Windows 11, Psychtoolbox 3.0.16, Psychopy, Internet
- **Personal PC:** Laptop etc

See the [Stimulus PC specification table](#) for full details

2) Connect the 'Master USB' to the PC

- By default, this will be connected to the WIN11 PC
- Connect 'Master USB' to any USB port of the desired PC
- This will connect: Mouse, Keyboard, Audio and Keypads



Standard Configuration of 7T Stimulus PC

3) Connect Projector

- **WIN11 STIM-PC:** Connected by default
- **Personal PC:** Connect HDMI labelled ‘Projector’ and connect to Personal PC

See the [Projector](#) page for full specifications

4) Turn ON Projector

HDBaseT Unit

- Press: **ON**
- LED will turn Green

On Remote

- Press: **ON**
- Wait for projector to turn on
- Epson Logo appears on small Siemens display



5) Select Projector Input

On Remote

- Press: **HDMI** or **HDBaseT**

WIN11 STIM-PC = **HDMI**

Personal PC = **HDBaseT**



6) Configure Scan Triggers

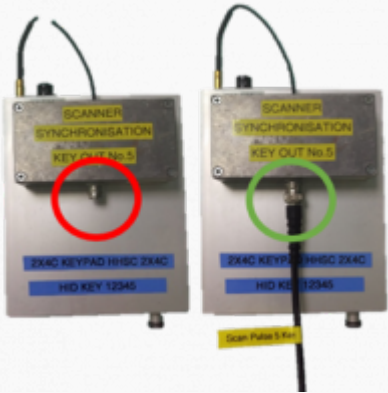
Key Press Triggers

- Trigger a script from a keypress of ‘5’
- Plug BNC ‘Scan Pulse 5’ into the keypad interface

Serial Triggers & No Triggers

- Trigger a script from serial output
- Unplug BNC ‘Scan Pulse 5’ from keypad interface
- *Personal Laptops will require a serial adapter, speak to cognitive interface*

See the [MRI Scan Synchronisation](#) page for full detail.



7) Turn ON Stimulus PC

Computer issues are reduced by connecting all devices and configuring the projector prior to boot

Standard Configuration of 7T Stimulus PC

8) Check Display Options

Display options can be configured but must be returned to default:

- Right Click Desktop
- Click: Screen Resolution/Display Settings

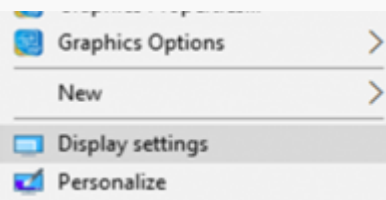
Default

Multiple displays: Extended these displays

Dell P2419H monitor : 1920×1080

(Recommended)

Epson PJ projector : 1920×1200 (Recommended)



When Finished

9) Turn OFF Projector

On Remote:

- Press: **Standby**, two times
- Wait for projector to turn OFF



10) Return To Default

- Revert any display settings back to default
- Reconnect any cables back to the WIN11 STIM-PC

11) Shutdown PC

MEG

Within the MEG suite a WIN11 STIM-PC is available.

The basic configuration of the system is shown below:

[MEG STIM-PC User Guide](#)

Standard Configuration of MEG Stimulus PC

1) Choose the Stimulus PC Required (*do not switch ON*)

- **WIN11 STIM-PC:** Windows 11, Psychtoolbox 3.0.16, Psychopy, Internet
- **Personal PC:** Laptop etc

See the [Stimulus PC specification table](#) for full details

2) Connect the 'Master USB' to the PC

- By default, this will be connected to the WIN11 PC
- Connect 'Master USB' to any USB port of the desired PC
- This will connect: Mouse, Keyboard, Audio and Keypads



Standard Configuration of MEG Stimulus PC

3) Connect Projector

- **WIN11 STIM-PC:** Connected by default
- **Personal PC:** Find cabl labelled 'Projector', this should not normally be connected, connect to personal PC

4) Turn ON Projector

- * Switch ON monitor "Power 1"
- * Switch ON interface "Power 2"
- * Switch ON projector "Power 3"
- * Press ON on the remote "Power 4"



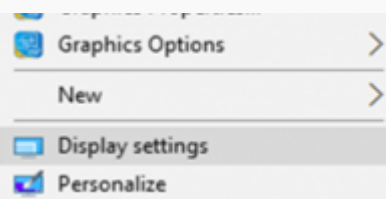
5) Turn ON Stimulus PC

Computer issues are reduced by connecting all devices and configuring the projector prior to boot

6) Check Display Options

Display options can be configured but must be returned to default:

- Right Click Desktop
- Click: Screen Resolution/Display Settings



Default

Multiple displays: Extended these displays

When Finished

7) Turn OFF Projector

- On Remote press OFF
- Wait for projector to turn OFF
- Switch OFF projector at "Power 3"
- Switch OFF PROpixx at "Power 2"
- Switch OFF LCD monitor at "Power 1"



8) Return To Default

- Revert any display settings back to default
- Reconnect any cables back to the WIN11 STIM-PC

9) Shutdown PC

IT Notes

Please Note:

- FIL IT credentials are required to log-on to FIL STIM-PCs.
- The Stimulus PCs are **NOT backed up!**

Please backup all scripts, data and log files.

- Please remove your data and scripts etc, when you've finished a study.
This ensures there is disk space for others to use.
- If you need to add one of your own toolboxes to the Matlab path;
Use the **addpath** command at the beginning of your script. Don't attempt to make a permanent change to the Matlab path variable, it can have (*and has had*) unpredictable effects on other fellows' studies.
- For a consistent user experience all STIM-PCs are maintained with the **same** set of software and toolboxes. These are frozen for long periods to ensure that long term studies have *consistent* results. Currently there is no date scheduled to update software versions from those listed in the [Specifications](#).
- The Toolboxes are limited (see [Specifications](#)) to get better Matlab startup/shutdown behaviour, partly to allow scripts to be reasonably portable to other scanning centres where Matlab licensing is per toolbox and expensive.
- The reliability of stimulus timing has been tested and verified with the above set of installed software and toolboxes. For this, and the previous note about consistency between STIM-PCs, **do not** expect to be able to install other 3rd party software or toolboxes on these PCs.

STIM-PC Specification

Please note that as of 02/09/2022 COGENT is no longer supported.

Current COGENT users will be supported until the project end however new projects will not be supported with COGENT.

	WIN11 STIM-PC
PC Model	Dell Optiplex 3070
Operating System	Windows 11 64-bit
Matlab Version	Matlab 2019a
Matlab Toolboxes	Data Acquisition Toolbox - for use with National Instruments (NI) data acquisition cards to record grip strength etc. Image Processing Toolbox Instrument Control Toolbox - for TCP/IP connections e.g. Brain Products fMRI EEG system. Signal Processing Toolbox - for image or sound file re-sampling and filtering. Statistics and Machine Learning Toolbox
Stimulus Packages	Psychtoolbox 3.0.16 Psychopy 2022.2.5
MRI Scan Synchronisation	Keypress '5' Serial input
Internet Connection	Yes
DVD/CD ROM	No
FIL Network Access	Full Access ABBA, ASIA, CHER .etc
Monitor	In most instances a Dell UltraSharp U2415 is in use These have a native resolution of 1920×1200 @60Hz

If these specifications do not meet your experimental requirements, we recommend using a [personal](#)

[device](#).

Stimulus Packages

For a consistent user experience all STIM-PCs are maintained with the same set of software and toolboxes. These are frozen for long periods to ensure that long term studies have consistent results. Currently there is no date scheduled to update software versions.

Psychtoolbox

Psychtoolbox 3.0.16 is installed on all WIN11 STIM-PC's.

Installation instructions for Fellow's PC's can be seen on <http://psychtoolbox.org/>.

The [WCHN Github](#) hosts examples Psychtoolbox scripts of common scripting tasks.

Psychopy

Psychopy 2022.2.5 is installed on all WIN11 STIM-PCs.

PsychoPy 2022.2.5 has some ability run scripts from older versions. See this link for details:

<https://psychopy.org/faqs/versionControl.html>

Installation instructions for Fellow's PC's can be seen on <https://www.psychopy.org/>.

Cogent2000

As of 02/09/2022 Cogent2000 is no longer supported.

Presentation

An [alternative stimulus presentation package](#) can be installed if required, and USB licenses keys are available. Please ask.

Example Stimulus Scripts

The [WCHN Github](#) hosts a series of Psychtoolbox example scripts that should be used for best case examples of common paradigm tasks.

Personal STIM-PC Use

It is possible to use a personal device as a Stimulus PC in the imaging suites.

Please note the following device requirements:

- Graphics output for [Projector](#) connection
 - HDMI port
 - *Display adapters are a researcher responsibility*
- 1 USB-A port for Master USB connection
 - Edirol UA-4FX sound card
 - [Keypad](#) input
 - MRI scanner synchronization
 - *Port adapters are a researcher responsibility*
- 1 USB-A port for each additional input
 - [Audiofile](#) sound card connection
 - USB/Serial adapter (*if serial [MRI scanner synchronization](#) is required*)
- Direct connection to the [Eyelink Eyetracker](#) is not available
 - Eyetracking data can be saved via the Eyelink PC

Windows Laptops

Laptops usually do not have a parallel port. This means parallel triggers, available in MRI and MEG, cannot be used.

Apple Mac Devices

Apple devices can specify the particular keyboard device from which to expect an input. This can cause issues if the ID of keyboard devices changes following a reboot etc.

If transferring a PsychToolbox script from Mac/Linux to Windows note that separate keyboards are NOT distinguishable in Windows. See [PTB KbCheck](#) :

“On Microsoft Windows, KbCheck can not address different keyboard devices, so selection by ‘deviceNumber’ only really works on Linux and macOS.”

Apple devices do not have parallel ports, neither is the io64 port driver available for Mac. This means the parallel triggers, available in MRI and MEG, cannot be used.

Thus in MEG we strongly recommend using the supplied Windows11 Stimulus PC with either PsychToolbox or PsychoPy to take advantage of the existing parallel port synchronisation triggers.

From:

<http://intranet.fil.ion.ucl.ac.uk/wiki/cogentwiki/> - **Cognitive Interface Wiki**

Permanent link:

http://intranet.fil.ion.ucl.ac.uk/wiki/cogentwiki/doku.php?id=stimulus_pc

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